

The Packing Number of Cubic Graphs

Wayne Goddard^{1,2} and Michael A. Henning²

¹ School of Mathematical and Statistical Sciences, Clemson University

² Department of Mathematical and Applied Mathematics, University of
Johannesburg

Abstract

A packing in a graph is a set of vertices that are mutually distance at least 3 apart. By using optimization and linear programming to help analyze the greedy algorithm, we improve on a result of Favaron and show that every connected cubic graph of order n has a packing of size at least $\frac{17}{132}n - O(1)$.

1 Introduction

Recall that a packing (or sometimes 2-packing) of a graph G is a set of vertices that are mutually distance at least 3 apart. The packing number of G , denoted $\rho(G)$, is the maximum size of a packing. In this paper we provide an improved lower bound on the packing number of a cubic graph. The seminal bound was provided by Favaron [4] in 1996, who showed that, except for the Petersen graph, the packing number of a connected cubic graph is at least $n/8$, where n is the order. In [8] it was noted that there are two cubic graphs of order 8 with diameter 2, and hence equality in the bound.

Favaron observed that in any connected cubic graph, the greedy algorithm of always taking a vertex at distance exactly three from the existing set, produces a packing of size at least $\lceil (n-2)/8 \rceil$. She then showed that, except for the Petersen graph, one can replace $n-2$ by n . More recently Bai et al. [1] showed that if the girth is 5 then the n can be replaced by $n+1$. Favaron's observation also immediately shows that every connected r -regular graph has a packing of size at least $\lceil (n-2)/(r^2-1) \rceil$. In [8] it was shown that one can replace the $n-2$ by n except when G has diameter 2 and order r^2+1 (that is, a Moore graph).

For some subfamilies of cubic graphs, the lower bound can be improved. For example, the packing number of a graph equals the independence number of its square; if a cubic graph has girth at least 6, then its square is 9-regular and

contains no K_5 , and by a result of Fajtlowicz [3] (see also [9]) has independence number at least $2n/(9 + 5) = n/7$. Recently, Gómez and Gutiérrez [7] showed that every bipartite cubic graph has packing number at least $7n/48$ for $n \geq 16$. For a related parameter, Balister et al. [2] showed that the 2-limited packing number of a cubic graph is at least $n/3$ (where a 2-limited packing is a set S such that every closed neighborhood contains at most 2 vertices of S —see [5]), and pointed out that this is best possible because of $K_{3,3}$.

We prove:

Theorem 1 *There is $B > \frac{1}{8}$ such that the packing number of every connected cubic graph G on n vertices is more than $B(n - 3)$.*

The proof is given in the next section and yields $B = 17/132 \approx 0.1287$. It seems unlikely that this value of B is best possible. In another direction, Favaron’s motivation was to prove upper bounds on the signed domination number of regular graphs. In particular, she noted that the signed domination number of a cubic graph equals $n - 2\rho(G)$. Thus the improved bounds for packing number also represent improved asymptotic upper bounds for the signed domination number. In [6] we considered the question of coloring the vertices of a graph without a rainbow P_3 , and noted that there is always a coloring with at least $\rho(G) + 1$ colors; thus Theorem 1 improves the lower bound for cubic graphs given in that paper.

The smallest asymptotic value of the packing number we can find is $n/7$. Here is a family of connected cubic graphs with packing number $n/7$. Let H be the subcubic graph of order 7 produced from K_4 by subdividing the three edges incident with one vertex. Take multiple copies of H and join them up to make a 3-regular graph. Since H has diameter 2, a packing of the resulting graph contains at most one vertex from each copy of H . And taking one vertex from each triangle produces a packing.

2 Proof of Theorem 1

We explain first the overall approach. The main idea is to produce a packing P by taking one vertex at a time. We say that a vertex is *live* if it is eligible

to be added to P , that is, distance more than two from every existing vertex of P ; otherwise it is **dead**. Favaron [4] observed that, while the initial vertex can cause up to ten vertices to become dead, if thereafter one takes at each step a live vertex v that has a dead neighbor, at most eight vertices become dead at each step. Thus at the end of the process when all vertices are dead, one has $8|P| + 2 \geq n$. To improve this we show that a positive fraction of the time one can choose a vertex where at most seven vertices become dead.

The key idea is to examine the situation when every possible live vertex makes at least eight vertices dead. One possible problem is that in the subgraph induced by the live vertices there is a component of order eight where every pair of vertices in the component have distance at most two in G . We call such a component **bad**. In Section 2.1 we show how to start with a good partial packing that precludes bad components in the future. In Section 2.2 we show that if bad components are not possible, then one can extend any partial packing adding at most eight dead vertices each time but where a positive fraction of the vertices taken add at most seven dead vertices.

2.1 Eliminating Bad Components

We consider here the subgraphs that can lead to bad components in the live graph. Define M_1 as the graph obtained from the Mobius ladder of order 8 (also known as the Wagner graph) by subdividing one edge. Define M_2 as the graph obtained from the Mobius ladder of order 8 by subdividing two edges as far apart as possible (equivalently, removing one edge from the Petersen graph). They are drawn in Figure 1.

For a set P of vertices let $N_2[P]$ be the set of all vertices within distance two of P .

Lemma 2 *For connected cubic graph G that is not the Petersen graph, there exists a partial packing P_0 such that $|N_2[P_0]| \leq \frac{38}{5}|P_0|$ and $N_2[P_0]$ contains all vertices from all copies of M_1 and M_2 except possibly the subdivision vertices.*

PROOF. If the graph contains no copies of M_1 or M_2 then P_0 being the empty set suffices. So assume this is not the case. We show first that the copies are

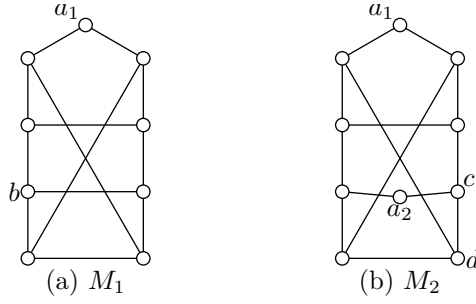


Figure 1: Subgraphs that can cause bad components

disjoint. In the case of M_1 , the third edge incident with the subdivision vertex is a bridge. Consider a copy of M_2 . Note that each edge is in a 5-cycle. Since we are not in the Petersen graph, the subdivision vertices are nonadjacent in G ; thus a cycle containing their third edge has length at least 6. In summary, if we were to remove all edges from G that are in neither a 4- nor a 5-cycle, the copies of M_i would remain as isolated components.

Let A_1 be the set of subdivision vertices in copies of M_1 , and A_2 be the set of subdivision vertices in copies of M_2 . Assume we have m_1 copies of M_1 and m_2 copies of M_2 . Then $|A_1| = m_1$ and $|A_2| = 2m_2$.

Construct from G an auxiliary graph F as follows. For each copy of M_1 , delete all vertices except the subdivision vertex. For each copy of M_2 , delete all vertices except the pair of subdivision vertices, and add an edge joining the pair. Note that distances in F are at most what they are in G .

Now, in F , vertices of A_1 have degree 1 and vertices of A_2 have degree 2. Indeed, if $a \in A_1$ then $N_2[a]$ contains at most three vertices of $A_1 \cup A_2$, while if $a \in A_2$ then $N_2[a]$ contains at most five vertices of $A_1 \cup A_2$. Let Q be a maximal subset of $A_1 \cup A_2$ that is a packing in F . Let p_1 and p_2 denote the number of vertices in Q corresponding to A_1 and A_2 respectively. By the maximality of Q , we have

$$3p_1 + 5p_2 \geq m_1 + 2m_2.$$

Note that, by the construction of F , each vertex of Q is from a different copy of M_i . In particular, $p_1 \leq m_1$ and $p_2 \leq m_2$.

Consider now Q as a subset of $V(G)$ and extend Q as follows. (i) For each M_1 ,

add inside M_1 a vertex b at distance 3 from a (see Figure 1(a)). (ii) For each $a \in Q \cap A_2$, add inside the corresponding M_2 one neighbor c of the other subdivision vertex (see Figure 1(b)). (iii) For each M_2 where neither subdivision vertex is in Q , add inside the M_2 one vertex d that is distance-two from both subdivision vertices. Let P_0 be the resultant set. Note that:

$$|P_0| = m_1 + m_2 + p_1 + p_2.$$

We claim that P_0 is a packing. Clearly operations (i) and (iii) preserve the packing condition. For operation (ii), in F there is no other vertex of Q within distance 2 of a , and hence no neighbor of the other subdivision vertex is in Q . So this operation preserves the packing condition.

We next bound $N[P_0]$. It can be checked that: For each $a \in Q \cap A_1$ there are at most 12 vertices within distance two of $\{a, b\}$; for each $a \in A_1 - Q$ there are 8 vertices within distance two of b ; and for each $a \in Q \cap A_2$ there are at most 14 vertices within distance two of $\{a, c\}$.

Consider some M_2 without a vertex of Q ; say vertex d was added when forming P_0 . We claim that at least two vertices of the M_2 are in $N_2[P_0 - \{d\}]$. This is immediate if either subdivision vertex of M_2 has a neighbor in Q . So assume otherwise. Then each subdivision vertex a of the M_2 is distance exactly 2 from a vertex $q \in Q$ in F , say via vertex x . If edge xq is in G , then $a \in N_2[q]$. On the other hand, if edge xq was added in forming F , then x is the other subdivision vertex of a copy of M_2 containing q , and by operation (ii) has a neighbor c in P_0 . It follows that $a \in N_2[c]$ and the claim is proven. That is, adding d to Q added at most 8 vertices to $N_2[P]$. Thus

$$|N_2[P_0]| \leq 12p_1 + 14p_2 + 8(m_1 + m_2 - p_1 - p_2) = 8m_1 + 8m_2 + 4p_1 + 6p_2.$$

If we divide everything by $m_1 + m_2$, it remains to maximize the ratio $(8m_1 + 8m_2 + 4p_1 + 6p_2)/(m_1 + m_2 + p_1 + p_2)$ subject to $3p_1 + 5p_2 \geq m_1 + 2m_2$, $0 \leq p_1 \leq m_1$, $0 \leq p_2 \leq m_2$, and $m_1 + m_2 = 1$. As per Wolfram Alpha, this program has maximum $38/5$ achieved at $(m_1, m_2, p_1, p_2) = (\frac{3}{4}, \frac{1}{4}, 0, \frac{1}{4})$. QED

It seems likely that the constant $\frac{5}{38}$ in Lemma 2 can be improved with more careful analysis. However, since this is bigger than $\frac{17}{132}$, it would not change the current constant in Theorem 1.

2.2 Completing the Packing

We consider the process of adding vertices to the packing P . At each stage, we let D denote the current set of dead vertices (that is, $N_2[P]$). Note that the subgraph induced by D has no isolate. Further, we define a **boundary edge** as an edge with one end live and one end dead; the **boundary** is the set of boundary edges.

We call a live vertex v a **j -vertex** if the number of live vertices in $N_2[v]$ is j . We further use the notation j^+ -vertex and j^- -vertex to mean a vertex where the number of live vertices in $N_2[v]$ is at least j and at most j .

Lemma 3 *Assume every live vertex is an 8^+ -vertex. Then*

- (a) *Every dead vertex that has one live neighbor has two live neighbors.*
- (b) *Every live vertex has at most one dead neighbor.*
- (c) *If have adjacent live vertices, at most one has a dead neighbor.*
- (d) *If have live quadrangle, then no vertex of the quadrangle has a dead neighbor.*
- (e) *If have live triangle, then every vertex of the triangle is distance at least three from its nearest dead vertex.*

PROOF. (a,b,c) Consider a live vertex v that has a dead neighbor m . By the above observation, vertex m has a dead neighbor. In order for taking v to contribute 8 vertices to D , all other vertices in $N_2[v]$ must be live. In particular, v has only one dead neighbor, and m has only one dead neighbor. Further, each live neighbor of v has no dead neighbor.

(d) If v is in a quadrangle, then there are at most 9 vertices in $N_2[v]$. But if v has a dead neighbor m then m has a dead neighbor, and so at most seven vertices of $N_2[v]$ are live, a contradiction.

(e) If vertex v is in a triangle, then there are at most 8 vertices in $N_2[v]$. It follows that all of these must be alive, and so v is distance at least three from any dead vertex. QED

Next we have our structural result. For $i = 1, 2$ let M'_i denote the graph obtained from M_i by deleting the i subdivision vertices. We say a live copy of M'_i is **bad** if in G it is embedded in a copy of M_i where the subdivision vertices are dead.

Lemma 4 *If every live vertex is an 8^+ -vertex, and there is no live component that is a bad M'_1 or M'_2 , then either:*

- (i) *there is an 8-vertex that increases the boundary by at least 4; or*
- (ii) *there is an 8-vertex that creates a 7^- -vertex (specifically one with a dead neighbor that has two dead neighbors).*

PROOF. Consider a live vertex v with dead neighbor m . Say the other neighbors of v are w_1 and w_2 ; further, say the other neighbors of vertex w_1 are x_1 and x_2 , while the other neighbors of vertex w_2 are y_1 and y_2 . See Figure 2.

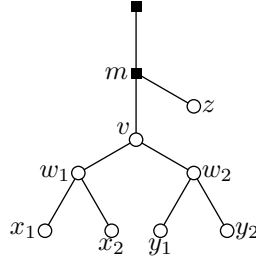


Figure 2: The 2-neighborhood of v

Let z be the other live neighbor of m . By Lemma 3, both neighbors of z other than m are live. If when v is taken one live neighbor of z lives and one dies, then the live one becomes a 7^- -vertex. So we may assume either both live neighbors of z are in $N_2[v]$ or neither is.

Case 1: *Both live neighbors of z are in $N_2[v]$.* By Lemma 3, vertex z is not in a live 4-cycle. So without loss of generality, it is adjacent to x_2 and y_2 . The third neighbor of x_2 cannot be dead, by Lemma 3 and the edge zx_2 . If the third neighbor is live but not in $N_2[v]$, then it becomes a 7^- -vertex when v is taken. So we may assume the third neighbor of x_2 is in $N_2[v]$. Similarly we may assume the third neighbor of y_2 is in $N_2[v]$. But x_2 and y_2 cannot be adjacent, since that creates a triangle that contradicts Lemma 3. Similarly, the edge x_1x_2 or y_1y_2 creates a triangle that contradicts the lemma. Hence there are edges x_2y_1 and x_1y_2 . See Figure 3.

By a similar argument, we may assume the third neighbor of x_1 is either dead or in $N_2[v]$, and similarly with the third neighbor of y_1 . If there is the edge x_1y_1 , then this is a bad copy of M'_1 , a contradiction. So assume the third neighbors of

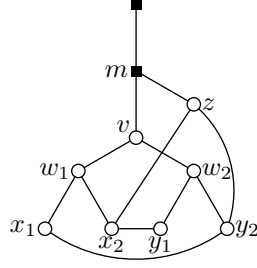


Figure 3: A possible subgraph

x_1 and y_1 are both dead. If they are the same vertex, then we have a bad copy of M'_2 , a contradiction. So assume the third neighbors are different vertices. Then the distance in G between x_1 and y_1 is three. It follows that if we take x_1 next, then y_1 remains live, and becomes a 2^- -vertex.

Case 2: z has no other neighbor in $N_2[v]$, and $N_2[v]$ does not contain a live cycle. By Lemma 3, each of x_1 and x_2 has at most one dead neighbor. If they both have a dead neighbor, then w_1 is currently a 7-vertex, a contradiction. So we may assume x_2 say has no dead neighbor. Similarly, we may assume that one of the neighbors of w_2 , say y_2 , has no dead neighbor. Thus the process of adding v to P changes at most four edges from boundary to not (one incident with each of v , z , x_1 and y_1) and changes at least eight edges from not to boundary (two incident with each of z , x_2 and y_2 , and one incident with x_1 and y_1). So there is a net increase of at least 4 in the boundary.

Case 3: z has no other neighbor in $N_2[v]$, and $N_2[v]$ contains a live cycle. As before neither of the edges x_1x_2 and y_1y_2 can be present as they would create a triangle that contradicted Lemma 3. Say there is edge x_2y_1 . See Figure 4. Consider the third neighbor of x_2 and of y_1 (possibly the same vertex). The third neighbor cannot be dead in both cases, since that contradicts Lemma 3. So say the third neighbor α of x_2 is alive. If α stays alive when v is taken, it will become a 7^- -vertex, since both of x_2 's other neighbors will become dead. So we may assume α is in $N_2[v]$.

So without loss of generality we may assume that vertex x_2 is adjacent to y_2 . That means there is a 4-cycle C containing x_2, y_2, y_1, w_2 . By Lemma 3, all other neighbors of C are live. But such a neighbor becomes a 7^- -vertex unless it too

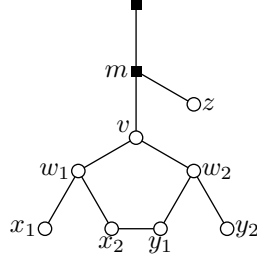


Figure 4: Another possible subgraph

dies. Since the edge y_1y_2 is precluded, it follows that we may assume both y_1 and y_2 are adjacent to x_1 . But then x_1 is currently a 7-vertex, a contradiction. QED

We next consider how in general the boundary can change. By “taking v ” we mean adding v to P .

- Lemma 5** (a) *If a live vertex v has three dead neighbors, then taking v increases D by at most 4 and decreases the boundary by at most 12.*
(b) *If no live vertex has three dead neighbors but there is one with two dead neighbors, then there exists a live vertex v such that taking v increases D by at most 6 and decreases the boundary by at most 8.*
(c) *If no live vertex has two dead neighbors, then taking any 7^- -vertex v decreases the boundary by at most 5.*
(d) *If every live vertex is an 8^+ -vertex, then taking any 8-vertex v decreases the boundary by at most 4.*

PROOF. (a) Assume vertex v has three dead neighbors. Since every dead vertex has a dead neighbor, it follows that $N_2[v]$ contains at most four live vertices. Each of the four vertices is incident with at most three boundary edges.

(b) Consider any vertex v with two dead neighbors and live neighbor w . Since every dead vertex has a dead neighbor, it follows that $N_2[v]$ contains at most six live vertices. See Figure 5.

Now, assume there is no edge connecting these six vertices, other than those incident with w . Then since vertices z_1 and z_2 each have at least one live neighbor, at worst they lose two boundary edges but gain one; and at worst the three

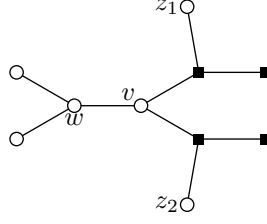


Figure 5: Another possible subgraph

neighbors of w lose two boundary edges each. This is a net boundary decrease of eight. One can then check that other cases have similar bounds. (Note that for example if z_1 and z_2 are adjacent and both have a second dead neighbor, then one can take one of them as the next v instead.)

(c) Let L_v be the subgraph induced by the live vertices in $N_2[v]$. Consider a vertex w in L_v . If w has degree 3 in L_v , then there is no boundary edge incident with w ; if w has degree 2 in L_v , then one boundary edge incident with w can be lost; and if w has degree 0 or 1, then at most one of its other neighbors is dead (by the hypothesis) and so the boundary incident with w cannot decrease. That is, the boundary decrease is at most the number of degree-2 vertices in L_v . It is easily checked that five is the maximum possible.

(d) By Lemma 3, every live vertex has at most one dead neighbor. So define L_v as in (c), and as before the decrease in the boundary is at most the number of degree-two vertices in L_v that have a dead neighbor. But by the same lemma, two such vertices cannot be adjacent. It is easily checked that four is the maximum possible in this case. QED

Finally we are in a position to prove Theorem 1:

PROOF. If the graph is the Petersen graph the result is immediate. So assume otherwise. If there is no copy of M_1 or M_2 , we start with P_0 an initial vertex. Else we start by applying Lemma 2 to yield P_0 . Let n' be the number of live vertices that remain after P_0 . In the first case $n' = n - 10$ and $|P_0| = 1 \geq \frac{5}{38}(n - 3 - n')$; and in the other case $|P_0| \geq \frac{5}{38}(n - n')$. At the end all of n' vertices are dead and no bad copy of M'_1 or M'_2 can occur.

Let a be the number of 3-dead-neighbor vertices taken, b the number of 2-dead-neighbor vertices taken, and c the number of other 7^- -vertices taken. Also

let d be the number of 8-vertices taken that create a 7^- -vertex, and e the number of 8-vertices of type (i) in Lemma 4. Then by Lemmas 4 and 5,

$$n' \leq 4a + 6b + 7c + 8d + 8e.$$

Further, since at the end the boundary is empty, the boundary constraint implies that

$$4e \leq 12a + 8b + 5c + 4d.$$

Since at each stage we always take one of the vertices that minimizes the increase in D , it follows that $a + b + c \geq d$.

So we minimize $a + b + c + d + e$ subject to these conditions. The linear program has optimum $17n'/132$ achieved at $c = d = n'/33$ and $e = 3n'/44$. That is $|P - P_0| \geq \frac{17}{132}n'$. QED

References

- [1] Xuqing Bai, Zhipeng Gao, Changqing Xi, and Jun Yue, On independent domination and packing numbers of subcubic graphs, arXiv:2401.04617, 2024.
- [2] P.N. Balister, B Bollobás, K Gunderson, Limited packings of closed neighbourhoods in graphs, arXiv:1501.01833, 2015.
- [3] S. Fajtlowicz, On the size of independent sets in graphs, Congr. Numer. 21 (1978), 269–274.
- [4] O. Favaron, Signed domination in regular graphs, Discrete Math. 158 (1996), 287– 293.
- [5] R. Gallant, G. Gunther, B.L. Hartnell and D.F. Rall, Limited packings in graphs, Discrete Appl. Math. 158 (2010), 1357–1364.
- [6] W. Goddard and H. Xu, Vertex colorings without rainbow subgraphs, Discuss. Math. Graph Theory 36 (2016), 989–1005.
- [7] Renzo Gómez and Juan Gutiérrez, Domination and packing in graphs, arXiv:2402.05088, 2024.

- [8] M.A. Henning and W.F. Klostermeyer, Packing in regular graphs, *Quaest. Math.* 41 (2018), 693–706.
- [9] M.A. Henning, C. Löwenstein, J. Southey, and A. Yeo, A new lower bound on the independence number of a graph and applications, *Electron. J. Combin.* 21 (2014), Paper 1.38, 12 pp.